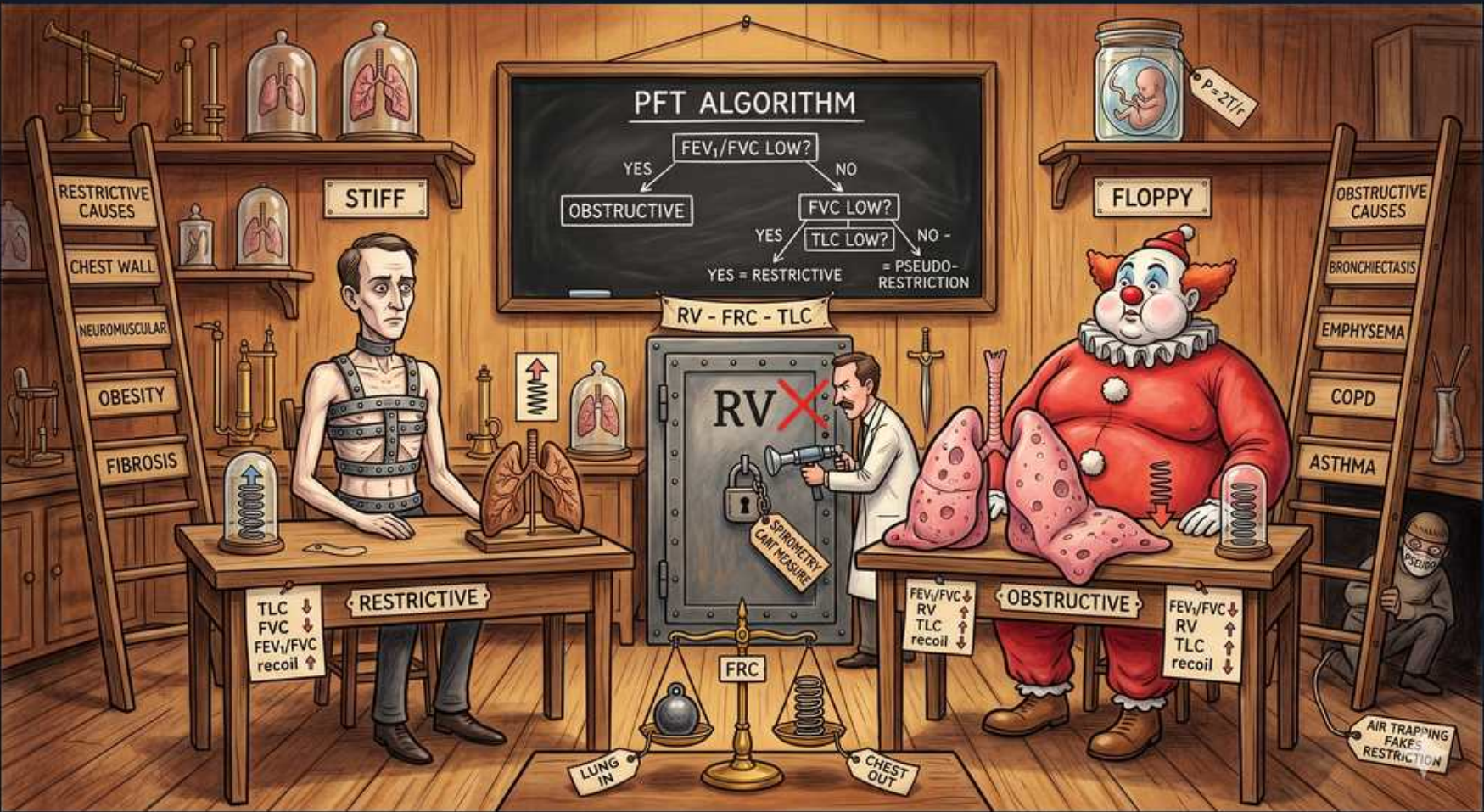


Respiratory Disturbance — PFT Workshop (Volumes & FEV₁/FVC Algorithm)



1. PFT algorithm board

WHAT YOU SEE

Blackboard flowchart starting 'FEV1/FVC low?'

WHAT IT ENCODES

PFT interpretation algorithm. Step 1: is FEV1/FVC < 0.7 (or below the lower limit of normal)? YES → OBSTRUCTIVE. If NO, step 2: is FVC low? YES → possible RESTRICTION (confirm with TLC); NO → normal. Step 3 for obstruction: assess severity by FEV1 % predicted and check bronchodilator reversibility.

BOARD TRAP

ALWAYS start with the FEV1/FVC RATIO, not the absolute FEV1 — a low FEV1 alone does not define obstruction (FEV1 falls in restriction too). The ratio is the first branch point.

2. Pseudo-restriction node

WHAT YOU SEE

Algorithm branch: TLC normal? No → pseudo-restriction

WHAT IT ENCODES

When FVC is low but the ratio is not obstructive, check TLC. NORMAL/high TLC with low FVC = PSEUDO-restriction (air trapping, submaximal effort, or coexisting obstruction), NOT true restriction. True restriction requires a LOW TLC.

BOARD TRAP

Spirometry alone CANNOT diagnose restriction — a low FVC only suggests it; you need lung volumes (TLC) to confirm. Low FVC + normal TLC = pseudo-restriction.

3. Restrictive causes shelf

WHAT YOU SEE

Left labels: chest wall, neuromuscular, obesity, fibrosis

WHAT IT ENCODES

Causes of restriction split into PARENCHYMAL (fibrosis/ILD) and EXTRApulmonary (chest-wall disease like kyphoscoliosis/ankylosing spondylitis, neuromuscular weakness, obesity, pleural disease).

BOARD TRAP

DLCO splits them: LOW DLCO in parenchymal fibrosis vs NORMAL DLCO in chest wall / neuromuscular / obesity. This is the single most useful next step after confirming low TLC.

4. STIFF (restrictive) man

WHAT YOU SEE

Thin man strapped/bound, labeled STIFF

WHAT IT ENCODES

Restrictive (fibrotic) lungs are STIFF: low compliance and INCREASED elastic recoil make them hard to expand, so all volumes shrink. The increased recoil is why the FEV1/FVC ratio is normal or high — airways are pulled open.

BOARD TRAP

Stiff/low-compliance lungs = restriction; do not confuse with the floppy, high-compliance lungs of emphysema (opposite end of the compliance spectrum).

5. RV–FRC–TLC safe

WHAT YOU SEE

Locked safe labeled RV-FRC-TLC with red X on RV

WHAT IT ENCODES

Lung volume definitions: TLC = total air at maximal inhalation; FRC = resting volume at end of a normal exhalation (ERV + RV); RV = air remaining after maximal exhalation. RV, FRC, and TLC all include RV.

BOARD TRAP

RV, FRC, and TLC CANNOT be measured by spirometry (because RV never leaves the lung) — they require body plethysmography or gas dilution (helium dilution / nitrogen washout).

6. FRC weights

WHAT YOU SEE

Balance scale labeled FRC, lung-in vs chest-out

WHAT IT ENCODES

FRC is the equilibrium point where the inward elastic recoil of the lung exactly balances the outward recoil of the chest wall (net zero pressure, no muscle effort). Emphysema (↓recoil) raises FRC; fibrosis (↑recoil) lowers it.

BOARD TRAP

Obesity lowers FRC and ERV FIRST (the outward chest-wall force is overwhelmed by mass) before other volumes change — the earliest obesity PFT abnormality.

7. RESTRICTIVE values

WHAT YOU SEE

Sign: TLC↓, RV↓, FEV1/FVC↑ (or normal), recoil↑

WHAT IT ENCODES

Restrictive pattern: ↓TLC, ↓FVC, ↓RV, with a NORMAL or HIGH FEV1/FVC ratio and increased elastic recoil. FEV1 is reduced but FVC falls proportionally more, keeping the ratio up.

BOARD TRAP

The FEV1/FVC ratio is NORMAL or HIGH in restriction — a LOW ratio rules restriction out (it indicates obstruction or a mixed pattern).

8. FLOPPY (obstructive) clown

WHAT YOU SEE

Big floppy clown labeled FLOPPY

WHAT IT ENCODES

Obstructive (emphysematous) lungs are FLOPPY: HIGH compliance and LOW elastic recoil. Loss of recoil lets airways collapse on expiration → air trapping → hyperinflation (↑RV, ↑TLC).

BOARD TRAP

Floppy/high-compliance = obstruction (emphysema), the exact opposite of stiff/low-compliance restriction. Compliance and recoil move in opposite directions in these two patterns.

9. Obstructive causes shelf

WHAT YOU SEE

Right labels: bronchiectasis, emphysema, COPD, asthma

WHAT IT ENCODES

Causes of obstruction (FEV1/FVC <0.7): asthma, COPD/emphysema, chronic bronchitis, and bronchiectasis. DLCO subdivides them: low in emphysema, normal/high in asthma and chronic bronchitis.

BOARD TRAP

Asthma obstruction REVERSES with bronchodilator (FEV1 rise ≥12% AND ≥200 mL); fixed COPD does not — reversibility testing is the discriminator.

10. OBSTRUCTIVE values

WHAT YOU SEE

Sign: FEV1/FVC↓, RV↑, TLC↑, recoil↓

WHAT IT ENCODES

Obstructive pattern: ↓FEV1/FVC (<0.7), ↑RV, ↑TLC (hyperinflation), ↑FRC, and reduced elastic recoil. A scooped/concave expiratory flow-volume limb accompanies it.

BOARD TRAP

An elevated RV/TLC in obstruction reflects air TRAPPING/hyperinflation — don't misread the high volumes as restriction (restriction LOWERS volumes).

11. Air trapping / fast restriction

WHAT YOU SEE

Bottom-right: faded figure, 'air trapping = fast restriction'

WHAT IT ENCODES

Severe obstruction with marked air trapping raises RV so much that FVC falls, mimicking restriction (pseudo-restriction). The distinguishing feature: TLC is HIGH in obstruction vs LOW in true restriction.

BOARD TRAP

A low FVC in an obstructive patient is pseudo-restriction from air trapping, NOT coexisting true restriction — confirm by measuring TLC (high = trapping, low = true restriction).

12. Lungs-in vs chest-out arrows

WHAT YOU SEE

Up/down recoil arrows beside the FRC scale

WHAT IT ENCODES

Elastic recoil determines resting volumes: LOSS of recoil (emphysema) shifts the balance outward → ↑FRC and ↑RV (hyperinflation); INCREASED recoil (fibrosis) pulls everything inward → ↓FRC, ↓RV, ↓TLC.

BOARD TRAP

Recoil direction predicts the volume change — but increased recoil ≠ obstruction. Increased recoil is the RESTRICTIVE (fibrotic) direction; decreased recoil is the obstructive (emphysematous) direction.